Phase changes

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Account for energy transfer during a phase change without a temperature rise.

SUCCESS CRITERIA - I can...

1. Identify a constant-temperature phase-change interval.
2. Use Q=mL for that interval.
3. Separate warming energy from phase-change energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “phase change” and “latent heat” to explain your answer. Sentence starters are optional.

phase change: a change between states such as solid and liquid

latent heat: energy per mass needed for a specified phase change

temperature: a measure of thermal state; it is not an energy amount

01 Warm-up: bring back an earlier idea

A sample warms by **3°C** after **500 J** is transferred into it. **Explain why the temperature rise** and heat transfer need different units.

Word help: heat = energy transferred because of a temperature difference

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

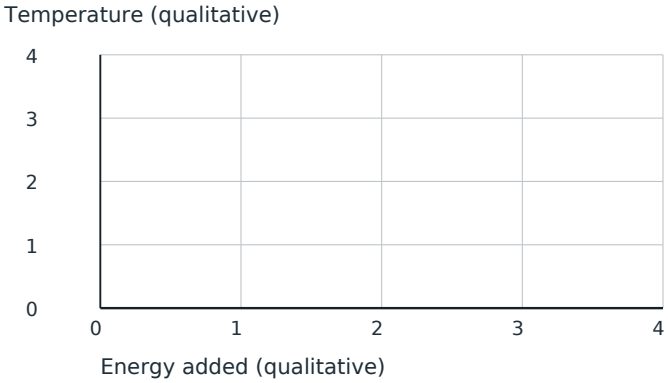
02 Worked example: explain the prediction

Reference: Q = mL; Q = mc ΔT.
Use mL during a phase change at constant temperature, and mc ΔT outside that interval. L is latent heat.

Worked reasoning: Melting **2 kg** of a substance with latent heat **100 J/kg** requires **200 J**. During ideal melting at fixed pressure, added energy changes phase without raising temperature.

Your turn: Sketch temperature versus added energy through warming and melting. **Mark the plateau**.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

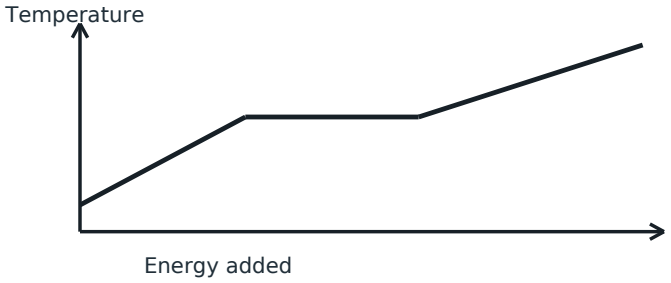


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

On the heating curve, name the flat interval. **Explain why added energy** does not raise temperature during that interval.

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .



The horizontal axis shows energy added. The vertical axis shows temperature.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A phase-change simulation uses latent heat L = **200 J/kg**. A **3 kg** sample starts at its transition temperature.
1. **Label the phase-change energy arrow** and **calculate the energy required** to finish the transition.
2. **Explain why this** does not find a temperature rise.

Sentence starter: I think … . I can point to …, which shows …, so … .

Initial T / phase: _____ Final T / phase: _____

Sample

Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student uses mc ΔT for a melting interval with ΔT=**0** and concludes no energy is required. **Explain the missing term**.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A spring has k=**400 N/m** and compression **0.06 m**. **Find force magnitude** and **describe its direction**. **Mark equilibrium**, displacement and restoring-force direction.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found … [unit]. This value means … in this situation.



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3

07 Exit check

A **2 kg** sample with L=**150 J/kg** changes phase completely.
1. **Find Q**.
2. **Label the energy arrow** and **identify what changes** in the sample.

Sentence starter: I found … [unit]. This value means … in this situation.

Initial T / phase: _____ Final T / phase: _____

Sample

Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Return to the worked example in problem **2**.
1. **Compare heat needed** to melt **1 kg** and **3 kg** of the same substance.
2. **Explain why temperature can remain** unchanged in both.
3. **Label the energy arrow** and **identify what changes** in the sample.

Sentence starter: Both … . They differ in … because … .

Initial T / phase: _____ Final T / phase: _____

Sample

Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: 1 2 3